In [2]:
```
import numpy as np
import pandas as pd

list1 = [[1, 'Smith', 50000], [2, 'Jones', 60000]]

df = pd.DataFrame(list1)
df.columns = ['EmpId', 'Name', 'Salary']
print(df)

df.info()

data = {
    'R&D Spend': [165349.20, 162597.70, 153441.51, 144372.41, 142107.34],
    'Administration': [136897.80, 151377.59, 101145.55, 118671.85, 91391.77],
    'Marketing Spend': [471784.10, 443898.53, 407934.54, 383199.62, 366168.42],
    'State': ['New York', 'California', 'Florida', 'New York', 'Florida'],
    'Profit': [192261.83, 191792.06, 191050.39, 182901.99, 166187.94]
}

df = pd.DataFrame(data)
df.info()
print(df.head())
print(df.tail())

employee_data = {
    'emp id': [1, 2, 3, 4, 5, 6, 7],
    'name': [
        'SREE VARSSINI K S',
        'SREEMATHI B',
        'SREYA G',
        'SREYASKARI MULLAPUDI',
        'SRI AKASH U G',
        'SRI HARSHAVARDHAN R',
        'GIRI HARSHAVARDHAN R'
    ],
    'salary': [5000, 6000, 7000, 5000, 8000, 3000, 6000]
}

df = pd.DataFrame(employee_data)

print(df.head())
print(df.tail())
df.info()

print(df.salary)
print(type(df.salary))
print(df.salary.mean())
print(df.salary.median())
print(df.salary.mode())
print(df.salary.var())
print(df.salary.std())

print(df.describe())
print(df.describe(include='all'))

empCol = df.columns
print(empCol)
```

```
emparray = df.values
print(emparray)

employee_DF = pd.DataFrame(emparray, columns=empCol)
print(employee_DF)
```

```
   EmpId   Name  Salary
0      1  Smith   50000
1      2  Jones   60000
<class 'pandas.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   EmpId   2 non-null      int64
 1   Name    2 non-null      str
 2   Salary  2 non-null      int64
dtypes: int64(2), str(1)
memory usage: 190.0 bytes
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   R&D Spend        5 non-null      float64
 1   Administration   5 non-null      float64
 2   Marketing Spend  5 non-null      float64
 3   State            5 non-null      str
 4   Profit           5 non-null      float64
dtypes: float64(4), str(1)
memory usage: 372.0 bytes
   R&D Spend  Administration  Marketing Spend       State     Profit
0  165349.20       136897.80        471784.10    New York  192261.83
1  162597.70       151377.59        443898.53  California  191792.06
2  153441.51       101145.55        407934.54     Florida  191050.39
3  144372.41       118671.85        383199.62    New York  182901.99
4  142107.34        91391.77        366168.42     Florida  166187.94
   R&D Spend  Administration  Marketing Spend       State     Profit
0  165349.20       136897.80        471784.10    New York  192261.83
1  162597.70       151377.59        443898.53  California  191792.06
2  153441.51       101145.55        407934.54     Florida  191050.39
3  144372.41       118671.85        383199.62    New York  182901.99
4  142107.34        91391.77        366168.42     Florida  166187.94
   emp id                 name  salary
0       1     SREE VARSSINI K S    5000
1       2           SREEMATHI B    6000
2       3               SREYA G    7000
3       4  SREYASKARI MULLAPUDI    5000
4       5         SRI AKASH U G    8000
   emp id                 name  salary
2       3               SREYA G    7000
3       4  SREYASKARI MULLAPUDI    5000
4       5         SRI AKASH U G    8000
5       6   SRI HARSHAVARDHAN R    3000
6       7  GIRI HARSHAVARDHAN R    6000
<class 'pandas.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   emp id  7 non-null      int64
 1   name    7 non-null      str
 2   salary  7 non-null      int64
dtypes: int64(2), str(1)
memory usage: 407.0 bytes
0    5000
```

```
1    6000
2    7000
3    5000
4    8000
5    3000
6    6000
Name: salary, dtype: int64
<class 'pandas.Series'>
5714.285714285715
6000.0
0    5000
1    6000
Name: salary, dtype: int64
2571428.5714285714
1603.5674514745463
         emp id       salary
count  7.000000     7.000000
mean   4.000000  5714.285714
std    2.160247  1603.567451
min    1.000000  3000.000000
25%    2.500000  5000.000000
50%    4.000000  6000.000000
75%    5.500000  6500.000000
max    7.000000  8000.000000
          emp id               name       salary
count   7.000000                  7     7.000000
unique       NaN                  7          NaN
top          NaN  SREE VARSSINI K S          NaN
freq         NaN                  1          NaN
mean    4.000000                NaN  5714.285714
std     2.160247                NaN  1603.567451
min     1.000000                NaN  3000.000000
25%     2.500000                NaN  5000.000000
50%     4.000000                NaN  6000.000000
75%     5.500000                NaN  6500.000000
max     7.000000                NaN  8000.000000
Index(['emp id', 'name', 'salary'], dtype='str')
[[1 'SREE VARSSINI K S' 5000]
 [2 'SREEMATHI B' 6000]
 [3 'SREYA G' 7000]
 [4 'SREYASKARI MULLAPUDI' 5000]
 [5 'SRI AKASH U G' 8000]
 [6 'SRI HARSHAVARDHAN R' 3000]
 [7 'GIRI HARSHAVARDHAN R' 6000]]
  emp id                  name salary
0      1     SREE VARSSINI K S   5000
1      2           SREEMATHI B   6000
2      3               SREYA G   7000
3      4  SREYASKARI MULLAPUDI   5000
4      5         SRI AKASH U G   8000
5      6   SRI HARSHAVARDHAN R   3000
6      7  GIRI HARSHAVARDHAN R   6000
```

In [ ]: